

Chapter 2 Logic of Quantified Statements

Predicate

Definition 2.1

A sentence is a **predicate** (or **propositional function**) if

- (1) it contains a finite number of **predicate variables** (eg: $x_1, x_2, \dots, x_n$), and
- (2) it is not a statement, but it becomes a statement when particular values are substituted for the variables.

The **universe of discourse** or **domain** D of a predicate variable is the set of all values that may be substituted in place of the variable.

Example 2.1

- (a) $P(x)$: x is an animal.

Domain: {apple, cat, computer, elephant, horse}

$P(\text{cat})$, $P(\text{elephant})$ and $P(\text{horse})$ are _____ statements.

$P(\text{apple})$ and $P(\text{computer})$ are _____ statements.

- (b) $Q(x)$: x is a positive integer. Domain: $\mathbb{R}$

$Q(2)$, $Q(100)$ and $Q(10,000)$ are _____ statements.

$Q(-10)$, $Q(0)$, $Q(\frac{1}{4})$ and $Q(1.15)$ are _____ statements.

- (c) $R(x, y)$: $x \geq y$.

The domain of $R(x, y)$ consists of pairs x and y where $x = 1, 2$, or 3 and $y = 3, 4$.

$R(3, 3)$ is a _____ statement.

$R(1, 3)$, $R(1, 4)$, $R(2, 3)$, $R(2, 4)$, $R(3, 4)$ are _____ statements.

Example 2.2

- (a) Let $P(x)$: x is prime $\Rightarrow x$ is odd. Indicate which of these statements are true and which are false.

(i) $P(13)$

(ii) $P(2)$

(iii) $P(5)$

(iv) $P(0)$

- (b) Let $P(x, y)$: $x^2 - y^2 = 0 \Leftrightarrow x = y$. What is the truth value of each of the following statements?

(i) $P(1, 1)$

(ii) $P(-1, 1)$

(iii) $P(1, 0)$

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Note

Symbol	Set
$\mathbb{R}$	Set of all real numbers
$\mathbb{Z}$	Set of all integers
$\mathbb{Q}$	Set of all rational numbers, or quotients of integers.

Definition 2.2

If $P(x)$ is a predicate and x has domain D , the **truth set** of $P(x)$ is the set of all elements of D that make $P(x)$ true when they are substituted for x . The truth set of $P(x)$ is denoted

$$\{x \in D \mid P(x)\}$$

which is read “the set of all x in D such that $P(x)$ ”.

Example 2.3

Let $Q(n)$ be the predicate “ n is a factor of 8.” Find the truth set of $Q(n)$ if

(a) the domain of n is the set $\mathbb{Z}^+$ of all positive integers.

(b) the domain of n is the set $\mathbb{Z}$ of all integers.

The Universal Quantifier: $\forall$

Definition 2.3

(1) Let $P(x)$ be a predicate and D the domain of x . A **universal statement** is a statement of the form

$$“\forall x \in D, P(x)”$$

means “ $P(x)$ is true for all values of x in D ”.

(2) It is defined to be **true** if, and only if, $P(x)$ is **true for every** x in D . It is defined to be **false** if, and only if, $P(x)$ is **false for at least one** x in D .

(3) A value for x for which $P(x)$ is false is called a **counterexample** to the universal statement.

Note

$\forall$ denotes “for all” and is called the **universal quantifier**.

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Example 2.4

Determine the truth value of the following universal statements. If the universal statement is false, suggest a counterexample to the universal statement.

(1) $\forall x \in \{-1, 0, 3, 4, 5\}, 2x$ is an even integer.

(2) $\forall x \in \mathbb{Z}, x + 1 < 4$.

(3) $\forall x \in \{x \mid 0 \leq x \leq 1\}, x < 5x$

(4) $\forall x \in \mathbb{Z}^+, (x - 1)$ is nonnegative.

(5) $\forall x \in \mathbb{Z}, (x + 1)$ is nonnegative.

(6) $\forall x \in \mathbb{R}, x^2 + 1 = 0$

(7) $\forall x \in \mathbb{R}, x^3 = x^2 - 2$

The Existential Quantifier: $\exists$

Definition 2.4

- (1) Let $P(x)$ be a predicate and D the domain of x . An **existential statement** is a statement of the form

$$“\exists x \in D \text{ such that } P(x)”$$

means “there exists an element x in D such that $P(x)$ is true.”.

- (2) It is defined to be **true** if, and only if, $P(x)$ is **true for at least one** x in D .

It is **false** if, and only if, $P(x)$ is **false for all** x in D .

Note

- (1) $\exists$ is called **existential quantifier**.
(2) $\exists$ means for some, there exists, for at least one, there is a, there is at least one.

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Example 2.5

Determine the truth value of each of the following existential statement.

(1) $\exists x \in \mathbb{Z}$ such that a prime number x is an even number.

(2) $\exists x \in \mathbb{R}^{\text{nonneg}}$ such that $x^2 + 5x + 6 = 0$.

(3) $\exists x \in \mathbb{R}$ such that $x = x + 1$

(4) $\exists x \in \mathbb{R}, x^2 + 1 = 0$

(5) $\exists x \in \mathbb{R}, x^3 = x^2 - 2$

Formal Versus Informal Language

Translating from Formal to Informal Language

Example 2.6

Rewrite the following formal statements in a variety of equivalent but more informal ways. Do not use the symbol $\forall$ or $\exists$.

(a) $\forall x \in \mathbb{R}, x^2 \geq 0$.

All real numbers have nonnegative squares.

Every real number has a nonnegative square.

Any real number has a nonnegative square.

x has a nonnegative square, for each real number x .

The square of any real number is nonnegative.

(b) $\forall x \in \mathbb{R}, x^2 \neq -1$.

All real numbers have square not equal to -1 .

No real numbers have square equal to -1 .

(c) $\exists m \in \mathbb{Z}$ such that $m^2 = m$.

There is an integer whose square is equal to itself.

We can find at least one integer equal to its own square.

$m^2 = m$, for some integer m .

Some integer equals its own square.

Some integers equal their own squares.

Translating from Informal to Formal Language

Example 2.7

Rewrite each of the following statements formally. Use quantifiers and variables.

(a) All triangles have three sides.

(b) No dogs have wings.

(c) Some programs are structured.

Universal Conditional Statements

$\forall x$, if $P(x)$ then $Q(x)$

Example 2.8

Rewrite the following formal statement in a variety of informal ways. Do not use quantifiers or variables.

$\forall x \in \mathbb{R}$, if $x > 2$ then $x^2 > 4$.

If a real number is greater than 2 then its square is greater than 4.

Whenever a real number is greater than 2, its square is greater than 4.

The square of any real number that is greater than 2 is greater than 4.

The squares of all real numbers greater than 2 are greater than 4.

Example 2.9

Rewrite each of the following statements in the form

$\forall$ _____, if _____ then _____.

(a) If a real number is an integer, then it is a rational number.

(b) All bytes have eight bits.

(c) No fire trucks are green.

Negations of Quantified Statements

Theorem 2.1 Negation of a Universal Statement

The negation of a statement of the form

$$\forall x \in D, P(x)$$

is logically equivalent to a statement of the form

$$\exists x \in D \text{ such that } \sim P(x).$$

Symbolically,

$$\sim(\forall x \in D, P(x)) \equiv \exists x \in D \text{ such that } \sim P(x)$$

Theorem 2.2 Negation of a Existential Statement

The negation of a statement of the form

$$\exists x \in D \text{ such that } P(x)$$

is logically equivalent to a statement of the form

$$\forall x \in D, \sim P(x)$$

Symbolically,

$$\sim(\exists x \in D \text{ such that } P(x)) \equiv \forall x \in D, \sim P(x).$$

Example 2.10

Write formal negations for the following statements:

- (a) $\forall$ primes p , p is odd.
- (b) $\exists$ a triangle T such that the sum of the angles of T equals 200° .

Note

You need to exercise special care to avoid mistakes when writing negations of statements that are given informally. One way to avoid error is to rewrite the statement formally and take the negation using the formal rule.

Example 2.11

Rewrite the following statement formally. Then write formal and informal negations.

- (a) No politicians are honest.

Formal version:

Formal negation:

Informal negation:

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(b) Some computer hackers are over 40.

Formal version:

Formal negation:

Informal negation:

Negations of Universal Conditional Statements

$\sim(\forall x, P(x) \Rightarrow Q(x)) \equiv \exists x$ such that $P(x)$ and $\sim Q(x)$

Example 2.12

Write a formal negation for statement (a) and informal negation for statement (b).

(a) $\forall$ people, if p is blond then p has blue eyes.

(b) If a computer program has more than 100,000 lines, then it contains a bug.

Statements containing multiple quantifiers

A **multiply-quantified statement** is a statement that contains more than one quantifiers.

Note

It is important to note that the order of the quantifiers is important, unless all the quantifiers are universal quantifiers or all are existential quantifiers.

Quantifiers of two variables

Statements	When True?	When False?
$\forall x \forall y P(x, y)$ $\forall y \forall x P(x, y)$	$P(x, y)$ is true for every pair x, y .	There is a pair x, y for which $P(x, y)$ is false.
$\forall x \exists y P(x, y)$	For every x , there is a y for which $P(x, y)$ is true	There is an x such that $P(x, y)$ is false for every y .
$\exists x \forall y P(x, y)$	There is an x for which $P(x, y)$ is true for every y .	For every x there is a y for which $P(x, y)$ is false.
$\exists x \exists y P(x, y)$ $\exists y \exists x P(x, y)$	There is a pair x, y for which $P(x, y)$ is true.	$P(x, y)$ is false for every pair x, y .

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Example 2.13

Let $P(x, y)$ be the statement “ $x + y = y + x$.” What is the truth value of the quantification $\forall x \forall y P(x, y)$.

The quantification $\forall x \forall y P(x, y)$ denotes the statement

“For all real numbers x and for all real numbers y , $x + y = y + x$ ”.

Since $P(x, y)$ is true for all real numbers x and y , the statement $\forall x \forall y P(x, y)$ is true.

Example 2.14

Let $Q(x, y)$ denote “ $x + y = 0$ ”. What are the truth values of the quantifications $\exists y \forall x Q(x, y)$ and $\forall x \exists y Q(x, y)$?

$\exists y \forall x Q(x, y)$: there is a real number y such that for every real number x , $Q(x, y)$.

No matter what the value of y is chosen, there is only one value of x for which $x + y = 0$. Since there is no real number y such that $x + y = 0$ for all real numbers x , the statement is false.

$\forall x \exists y Q(x, y)$: for every real number x there is a real number y such that $Q(x, y)$.

Given a real number x , there is a real number y such that $x + y = 0$; namely, $y = -x$. Hence the statement is true.

Example 2.15

Rewrite the following statements formally using quantifiers and variables.

- (1) Given any positive number, there is another positive number that is smaller than the given number.
- (2) Any even integer equal twice some other integer.
- (3) Somebody trusts everybody.
- (4) For every real number x and every real number y , if x is positive and y is negative, then the product of x and y is negative.

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Example 2.16

Translate each of the following statements into English:

- (1) $\forall x \in D, (C(x) \vee \exists y \in D \text{ such that } (C(y) \wedge F(x, y)))$
where $C(x)$: x has a computer, $F(x, y)$: x and y are friends, and
 D : all students in UTAR.
- (2) $\forall x \in D, \exists y \in D \text{ such that } ((F(x) \wedge P(x) \Rightarrow M(x, y)))$
where $F(x)$: x is a female, $P(x)$: x is a parent, $M(x, y)$: x is the
mother of y and D : all human beings.

Example 2.17

Let $Q(x, y)$ be the statement “ $x + y = x - y$ ”. if the universe of discourse for both variables consists of all integers, what are the truth values?

- (a) $\forall y Q(1, y)$
- (b) $\exists x \exists y Q(x, y)$
- (c) $\forall x \exists y Q(x, y)$
- (d) $\exists y \forall x Q(x, y)$
- (e) $\forall y \exists x Q(x, y)$

Negations of Multiply-Quantified Statements

$\sim(\forall x \exists y \text{ such that } P(x, y)) \equiv \exists x \text{ such that } \forall y \sim P(x, y).$

$\sim(\exists x \text{ such that } \forall y P(x, y)) \equiv \forall x \exists y \text{ such that } \sim P(x, y).$